DEP SPECIFICATION

ROADS, PAVING, SURFACING, CABLE TRENCHES, SLOPE PROTECTION AND FENCING

DEP 34.13.20.31-Gen.

September 2011

DESIGN AND ENGINEERING PRACTICE

PREFACE

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1. INTRODUCTION

1.1 SCOPE

This DEP specifies requirements and gives recommendations for the design and construction of roads, paving, surfacing of unpaved areas, cable trenches, erosion protection of slopes, and fencing and gates.

The minimum technical requirements as laid down in this specification shall be applied. Supplementary to these requirements, the design and construction shall be in accordance with generally accepted theories, codes, methods and good working practices. Examples of acceptable codes are referenced in this DEP. The climate, topography, soil conditions and local requirements shall be taken into account as well as the specific requirements of the Principal. Equipment, materials and working methods shall, unless specified otherwise, be subject to approval by the Principal.

Requirements described herein relate to and are limited to work above the sub-grade. If a functional minimum property of the subgrade is required, that property will be stated without further discussion.

Geotechnical, foundation engineering and surfacing aspects, including the stability of slopes, roads, bund walls, tank pits and tank pads, are not within the scope of this DEP. For these aspects, reference is made to DEP 34.11.00.11-Gen. and DEP 34.11.00.12-Gen.

Relevant standard drawings are referenced in DEP 00.00.06.06-Gen., Index to Standard Drawings.

This DEP is a revision of the DEP of the same title and number dated April 2003. A summary of changes since the previous edition is given in (1.5).

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this DEP is confined to Shell companies and, where necessary, to Contractors and Manufacturers/Suppliers nominated by them. Any authorised access to DEPs does not for that reason constitute an authorization to any documents, data or information to which the DEPs may refer.

This DEP is intended for use in facilities related to oil refineries, chemical plants, gas plants, or any such facility related to manufacturing activities and, where applicable, in exploration, production and distribution. This DEP may also be applied in other similar facilities.

When DEPs are applied, a Management of Change (MOC) process should be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this DEP, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and which combination of requirements will be acceptable with regards to the safety, environmental, economic and legal aspects. In all cases the Contractor shall inform the Principal of any deviation from the requirements of this DEP which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned, the objective being to obtain agreement to follow this DEP as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

1.3.2 Specific definitions

Note: Refer to Figure 1.1 and Figure 2.1 for illustration of common terms.

Term	**Definition**
Site	An area comprising the total of all on-plot and off-plot areas, which may be located either onshore, nearshore, offshore, or at a combination of these locations.
Off-site	Areas associated with the site, but located at a distance, e.g., water wells.
On-plot area	An area designated for utility and processing units, including associated control rooms, electrical sub-stations, analyser houses, stacks, associated pipe tracks and plant roads.
Off-plot areas	An area designated for all facilities not in the on-plot area which may include administration buildings, workshops, laundries, warehouses, materials yards, storage tank compounds, pump stations, flares, fire-fighting station, training grounds, cooling water intake station, cooling towers, cooling water settling ponds, cooling water discharge channels, jetties, harbour, associated pipe tracks and roads.
Battery limit	The operational boundary of individual processing or utility units.
Sub-grade	The surface prepared to support a pavement system.
Sub-base	The engineered granular fill placed on the sub-grade with the purpose of supporting the bound materials of the pavement system.
Road width	The distance between the outside edges of the road pavement. This does not include the width of shoulders.

1.4 CROSS-REFERENCES

Where cross references to other parts of this DEP are made, the referenced section number is shown in brackets ( ). Other documents referenced in this DEP are listed in (9).

1.5 SUMMARY OF CHANGES SINCE PREVIOUS EDITION

The previous edition of this DEP was dated April 2003. Other than editorial and formatting revisions, the following are the main changes to that edition:

In this revision normative requirements have been summarised.

Old section	New Section	Change
2.1	2.1	Included reference to DEP 80.00.10.11-Gen.
2.4.3	2.4.3	Updated cross-fall to 1:50 to remove conflicts with drawings 13.007-A and 2.4.10.
	2.4.9	New section on road barriers.
3.4.5	3.4.5	Included section on brick paving.
3.5	3.5	Updated for thickness of polythene.
4.3	4.3	Updated surfacing requirements for unpaved areas.
5	5	Cable trench requirements updated for alignment with electrical DEPs.
7.2	7.2	Updated for Security Risk Assessment requirements
7.4.1	7.4.1	Updated fence functional requirements.

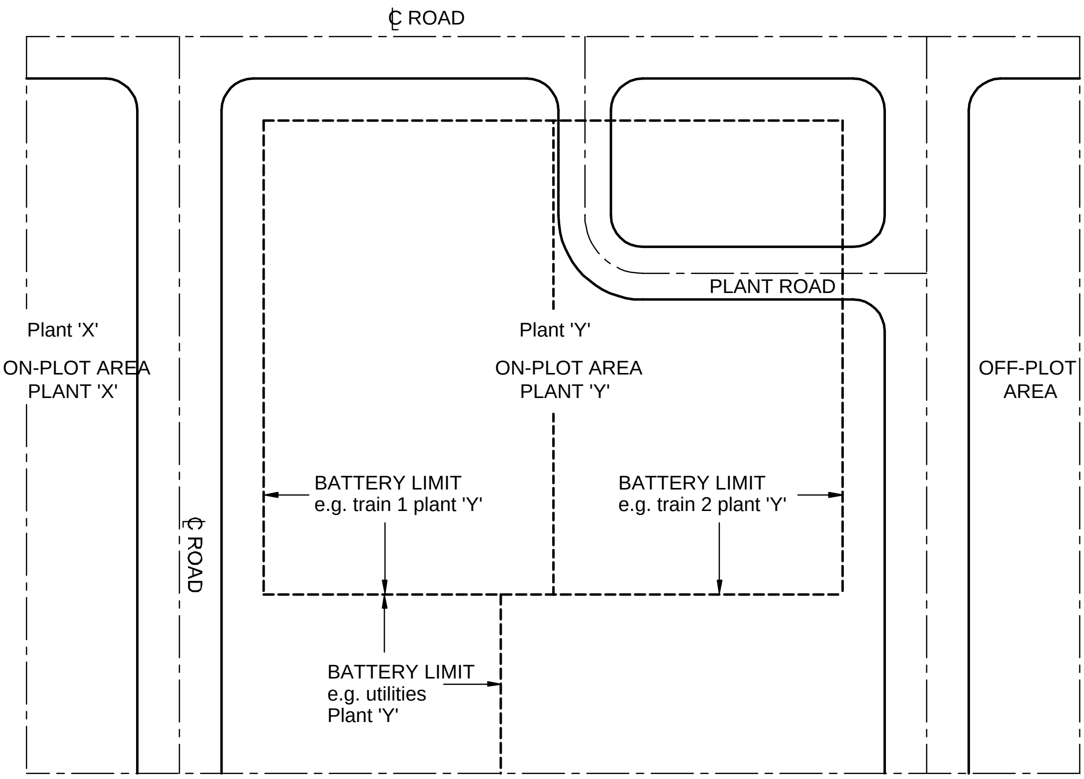


Figure 1.1 Schematic Diagram Showing Typical Terms Used

1.6 COMMENTS ON THIS DEP

Comments on this DEP may be sent to the Administrator at standards@shell.com, using the DEP Feedback Form. The DEP Feedback Form can be found on the main page of “DEPs on the Web”, available through the Global Technical Standards web portal http://swww.shell.com/standards and on the main page of the DEPs DVD-ROM.

1.7 DUAL UNITS

This DEP contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

2. ROADS

2.1 DETAILED SCOPE

This section outlines the minimum requirements for permanent roads and bridges within the site.

Unless otherwise specified by the Principal, all road widths shall as a minimum comply with the following table:

Road Type	Road Width (m)	Road Width (ft)	Notes
Main	8	26	shoulder
Plant	4 to 6	13 to 20	shoulder
off-plot heavy-duty	6	20	shoulder
off-plot light-duty	4	13	shoulder
Patrol	3	10	no shoulder
maintenance track	to suit	to suit	no shoulder

Roads shall be assigned in line with the plant layout strategy; reference is made to DEP 80.00.10.11-Gen.

General terms used to describe roads and pavements are illustrated in the cross sections presented in Figure 2.1.

The design of contaminated and non-contaminated drainage systems is not covered by this technical specification. The design of these drainage systems is described in DEP 34.14.20.31-Gen.

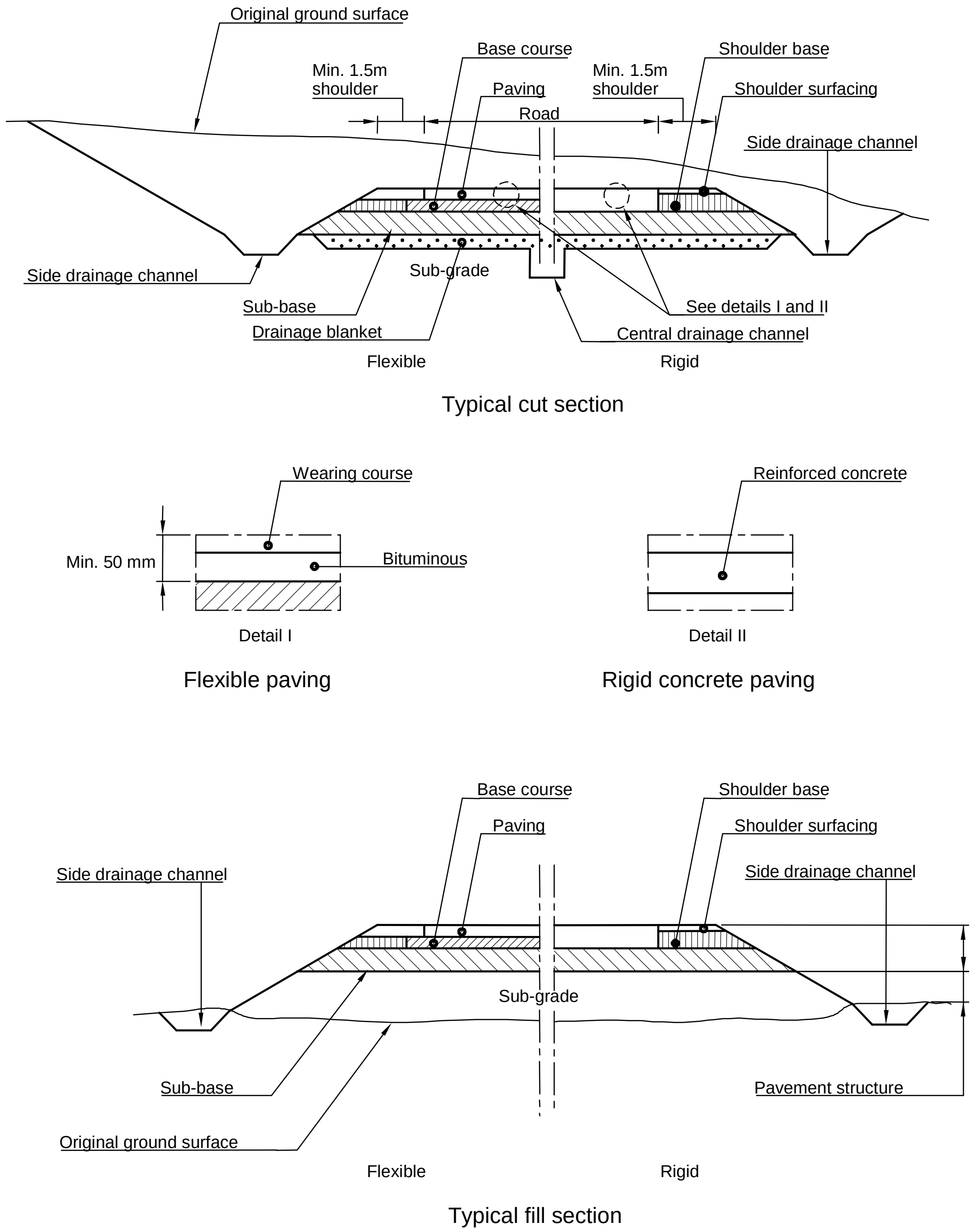


Figure 2.1 Schematic Drawing Showing Typical Cross Sections Of Roads

2.2 GENERAL DESCRIPTION

The road pavement structure shall be supported by the sub-grade and, unless specified by the Principal, shall comprise:

- sub-base;
- paving.

A sub-base is required if the settlement behaviour and/or the bearing capacity of the sub-grade are insufficient to support the paving and traffic without excessive deformations or maintenance during the lifetime of the plant. Where the support by the sub-grade is insufficient, consideration should be given to economic methods of ground improvement, the suitability of which shall be demonstrated by site trials. Details of ground improvement are provided in DEP 34.11.00.11-Gen.

Roads shall be of the two-lane type, except for maintenance tracks, patrol roads and other roads as specified by the Principal. Terms are defined and typical cross sections of roads are shown in Figure 2.1.

Shoulders shall be capable of supporting, without damage, slow-moving and parked vehicles for which the road is designed. In the design of roads, sufficient shoulder width shall be provided to accommodate cable trenches.

2.3 FUNCTIONAL REQUIREMENTS

The functional requirement for all roads and tracks is to provide:

- stable and durable access to all locations of the site for those transport means for which they are designed and may include cars, trucks, cranes and other mobile equipment such as snow ploughs, fire engines or road sweepers that are required at those locations during the lifetime of the plant;
- stable and durable access to all installations and buildings for fire-fighting equipment and other emergency vehicles under consideration for the site via at least two independent routings under all conditions; see DEP 80.47.10.33-Gen.

2.4 DESIGN AND MATERIALS

2.4.1 General

Roads on which a hydrocarbon or chemical spillage is expected during the lifetime of the road shall be surfaced with chemical-resistant paving and may be of a concrete type design.

In the selection of paving material, consideration shall be given to operational requirements during the life of the plant as well as the cost of construction and maintenance of the pavement. In some circumstances, there may be a requirement to install or access underlying services. In such situations, paving slabs, brickwork or blockwork may provide a suitable low maintenance flexible paving system.

Design, materials and testing procedures, etc., shall be in accordance with international standards and codes, as specified in this DEP.

2.4.2 Interface with site preparation

The design of the sub-grade shall be based on the result of a geotechnical (soil mechanical and geohydrological) survey, including calculations and determination of the grain size distribution, silt content, plasticity index, friction properties, relative density and permeability. The average, high and the highest possible ground water levels, as well as the in-situ permeability, shall also be assessed. The objective of these investigations is to ensure the sub-grade remains stable and durable throughout the life of the plant. Depending on the location of the site, or the processes for which the plant is designed, some sub-grade materials may be affected by swelling, shrinkage or softening due to changes in moisture content, heave due to frost action, or deterioration due to chemical reaction.

Acceptance criteria for the sub-grade and sub-base shall be developed on the basis of the design requirements of the pavement. Before surfacing, acceptance of the sub-grade and sub-base shall be obtained through compliance testing, the results of which shall meet the requirements of the acceptance criteria.

The design shall include measures to prevent intermixing of sub-grade material into the sub-base during construction and during the design life of the pavement.

2.4.3 Heavy-and light-duty roads

The pavement of heavy- and light-duty roads shall be designed in accordance with the latest revision of AASHTO GDPS-4 or equivalent standard as acceptable to the Principal.

Details of heavy and light duty roads and patrol roads are presented in Standard Drawings S13.005, S13.006 and S13.007.

Design factors that are prescribed by, but not quantified in, the above guide shall be proposed to and agreed with the Principal.

The total thickness shall be designed for the total number of axles indicated for the total design life. The recommended number of axle movements for the design of heavy-duty roads is 1.5 million and for light-duty roads 1.0 million.

The terminal serviceability index for heavy- and light-duty roads shall be 2.5 as defined in the AASHTO publication. The concrete working stress as defined in this publication may be derived from the standard splitting strength, see DEP 34.19.20.31-Gen.

In design, the following parameters shall be considered:

- reliability, R = 75 % for heavy duty roads;
- reliability, R = 50 % for light duty roads;
- standard deviation $S_o = 0.45$;
- initial serviceability index, $P_o = 4.2$.

The axle loads and arrangement of vehicle dimensions for heavy- and light-duty roads are defined in Figure 2.2.

The road construction, i.e., sub-base and paving, shall be adequately supported horizontally either by a sufficiently paved shoulder or by a vertical retaining element, e.g., concrete kerb.

The roads listed below shall be of the heavy-duty type:

- main roads on site and all roads in and around processing units, utility areas and yards;
- roads to and around main buildings and loading facilities;
- main access roads to the site whose construction and maintenance are not the responsibility of government authorities.

Light-duty roads comprise all other roads on site except:

- patrol roads (2.4.4);
- maintenance tracks (2.4.5).

Road widths shall be determined through consideration of service requirements during the life of the plant and shall comply with the minimum requirements of (2.1). Road shoulders on either side shall have a minimum width of 2 m (6 ft – 6 in) for heavy and 1.5 m (5 ft) for light-duty roads, unless otherwise specified by the Principal.

The boundary between the road and road shoulder shall be clearly marked with lines or a kerb. Openings shall ensure the de-watering of the road surface.

All heavy-duty and light-duty roads shall have a cross fall generally from the centre or crown of the road, but occasionally continuously across the road. The crossfall shall have a gradient of 1 in 50 to ensure effective run-off of rainwater. The paved areas adjacent to the roads shall have a smooth transition to the edge of the roads. The maximum gradient in the longitudinal direction of heavy-duty roads shall be 1 in 20 and for light duty roads 1 in 10.

Flexible pavings shall have a minimum thickness of 50 mm (2 in).

Rigid paving shall have a minimum thickness of 250 mm (10 in) for heavy-duty roads and 150 mm (6 in) for light-duty roads.

2.4.4 Patrol roads

Patrol roads shall be designed to accommodate vehicles with a maximum weight of 5 t (11,000 lbs) under all relevant climatic conditions. Minimum width shall be as specified in (2.1). The maximum gradient shall not exceed 1 in 8.5. Patrol roads shall be paved with a flexible-type paving or economic equivalent.

2.4.5 Maintenance tracks

Maintenance tracks specified by the Principal to facilitate access to future development areas within the site boundary shall be accessible for 4-wheel drive-type vehicles under normal weather conditions. Vegetation, peat, and soft clays shall be removed. The maximum gradient shall not exceed 1 in 5. Surfacing is not required, however the minimum requirements as specified in (2.4.2) shall be met.

2.4.6 Road crossings

Roads should cross at an angle of 90 degrees.

At road crossings, the edge of the paving shall have a radius of curvature of:

7.5 m (24.5 ft)	for 4 m (13 ft) wide roads;
10 m (33 ft)	for 6 m (20 ft) wide roads;
12 m (40 ft)	for roads with a width of 8 m (26 ft) or more.

For crossings of roads of unequal width, the narrower road shall determine the radius of the curvature.

The edges defining the radius of curvature at the junction of a heavy-duty road crossing shall be designed to be at the same level.

2.4.7 Bridges, culverts and pipe crossings underneath roads

In this DEP, design considerations for the above are limited to those that interface with road design. The following axle loads shall be allowed for:

- heavy-duty roads - A tandem axle load of 150 kN (33.7 kips) at the front of a car combined with an axle load of 150 kN (33.7 kips) at the rear; see Figure 2.2.
- light-duty roads - two axles of 50 kN (11.25 kips) each, two wheels at a distance of 2 m per axle and an axle to axle distance of 3 m (10 ft); see Figure 2.2.

The minimum distance between the finished road surface and the top of underlying pipelines, pipe bridges and other crossings shall be 0.5 m (1 ft - 8 in).

Special load requirements for specific current and anticipated future duties, including construction or maintenance requirements, shall be considered separately.

At least one road leading to the main process or distribution area(s) shall be designated as a heavy equipment route and bridges/culverts including other underground facilities shall be designed for the maximum expected loading condition caused by transportation of heavy equipment.

2.4.8 Crash barriers

Crash barriers shall be applied where considered essential to protect personnel, equipment and piping. Typical locations requiring crash barriers are road crossings over pipe tracks and under pipe racks, roads close to pipe racks and pipe tracks, and roads adjacent to steep slopes, T-junctions and sharp bends.

The preferred crash barrier system consists of I-beams with steel supports embedded in the soil at 3 m (10 ft) distances to carry a guard-rail of 0.75 m (2 ft - 6 in) minimum height. The steel should be galvanized.

The system shall be designed to absorb the design impact energy by deformation without collapse.

Details of design and materials for crash barriers shall be submitted to the Principal for review.

2.4.9 Road barriers

All road barriers shall be of the vertical swung type, either counterweighted or with integrated gas compression.

The material of the barrier arms should preferably be of aluminium to allow easy deformation, or in case of other material, should be provided with breaking pins.

Barrier arms shall not be locked or otherwise inhibit access by emergency teams. Arms shall be fully painted in red/white reflecting bands. The plant road and the barrier should be well lighted by lampposts.

The barriers should preferably consist of short (2 or 3 metres) arms without support at the end of the arm. There shall be an easy passage for pedestrians and cyclists, e.g. by creating an opening of 1 or 1.5 metres between the arms, or in case of one arm between the end of the arm and the edge of the road. The arms shall be free of any (traffic) signs to prevent extreme exposure to wind loads.

2.4.9 Exceptional transport on existing roads

The most severe conditions of either transport vehicles carrying exceptionally heavy equipment, or of vehicles with multiple axles providing continuous and high load, shall be adopted as the extreme loading conditions in the design of roads (including culverts and bridges). If these are not acceptable, an adequate solution shall be agreed with the Principal.

2.4.10 Drainage

2.4.10.1 Surface drainage

The road surface shall have a minimum lateral gradient of 1 in 50 to provide proper drainage under all weather conditions.

The drainage of road surfaces and plot paving forms an integral part of the drainage system. For its design, reference is made to DEP 34.14.20.31-Gen.

2.4.10.2 Control of ground water

The ground under the road paving system shall be well drained.

The distance between the underside of the paving and the normal high ground water level shall be not less than:

Road Type	Distance to water (m)	Distance to water (ft - in)
Main	0.75	2 ft - 6 in
Plant	0.50	1 ft - 8 in
off-plot heavy-duty	0.50	1 ft - 8 in
off-plot light-duty	0.50	1 ft - 8 in
Patrol	0.30	1 ft
maintenance track	not applicable	not applicable

If the sub-grade comprises rock, the normal high ground water level shall be taken as the sub-grade level, unless it is demonstrated to the satisfaction of the Principal that the normal high ground water level is below sub-grade level. This applies only when the distance between rock level and underside of paving is less than the above values.

A permanent sub-surface drainage system shall be designed to ensure control of the ground water level with regard to the above stated minimum distances and will include the drainage of the subsoil under and adjacent to the road.

The influence of unlined drainage ditches, lined drain ditches and channels with weep holes on the ground water table shall be taken into account. If there is a reasonable risk of hydrocarbon or chemical spillages entering the roadside drainage system, the system shall be lined to prevent contamination of the underlying ground. In such instances, the ground water levels beneath the road shall be controlled by an independent system which may include a buried drainage blanket and central drain as indicated in Figure 2.1, Typical cut section.

The drainage system shall be designed to remain effective for the life of the plant and its design shall prevent the unacceptable migration of fine soils into the drainage layers or surface ditches.

2.4.10.3 Roads crossing existing drainage or other piping systems

If the construction of roads may interfere with the existing surface drainage system, their design shall ensure the existing drainage system is unaffected. Measures taken to achieve this may include, but not be limited to, the installation of culverts and ditches and the re-routing of water courses; see DEP 34.14.20.31-Gen. The drainage system shall be designed on the basis of a storm with a recurrence period of 30 years, or the design life of the plant, whichever is the longer period. If available, the results of a hydrological survey shall be taken into account.

Surface water run-off from catchment areas outside the site shall be diverted around the site by means of perimeter ditches, which shall be designed to accommodate the peak design flows and protected against scour. The location and routing of the ditches shall be established in consultation with the Principal.

2.4.11 Transition slabs

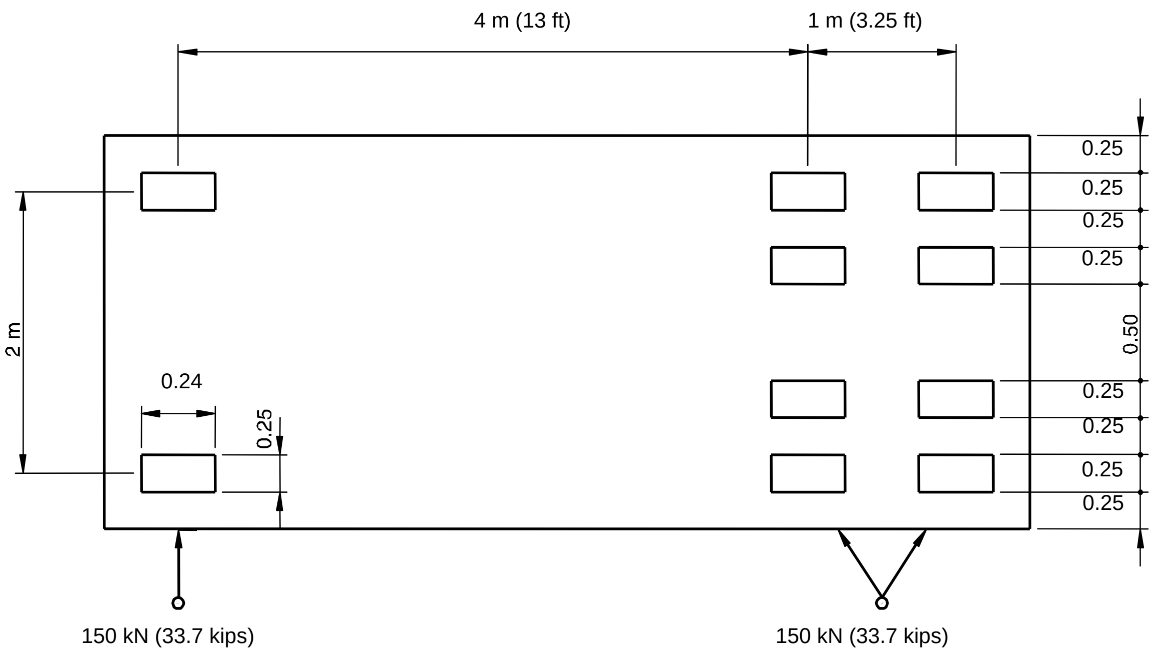
The use of below ground, reinforced concrete transition slabs shall be considered at those locations where differential settlement between paving and/or foundations is expected. This may be the case at pipe culverts and/or piled objects such as foundations and liquid-light process plant paving.

2.5 CONSTRUCTION

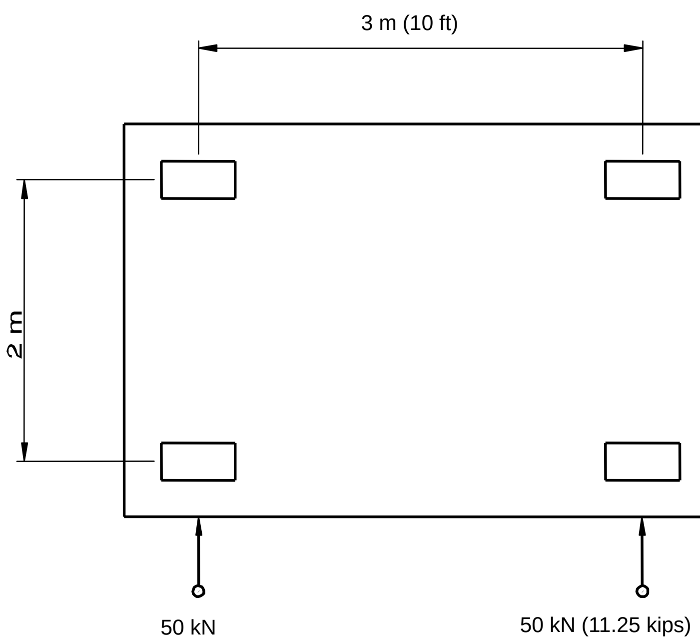
In the interest of safety, it shall be ensured that all parts of the site are accessible by emergency vehicles, unless permission is obtained from the Principal for temporary works.

During construction, a traffic management system shall be implemented to minimise the risk of road traffic accidents, injury to personnel and damage to or loss of nearby equipment.

Details of proposed materials for road construction and the compliance testing programme to control the works shall be submitted to the Principal for review.



On heavy-duty roads



On light-duty roads

Figure 2.2 Schematic Drawing Showing Typical Axle Loads And Vehicle Dimensions

3. PAVING

3.1 DETAILED SCOPE

This section describes the minimum requirements for paved areas. In general, these areas should be limited to those parts of a site where:

- there is a reasonable risk that there will be an accidental spillage from operations and maintenance of chemicals or hydrocarbons which would be detrimental to the environment;
- where design loads other than pedestrians are to be expected.

3.2 GENERAL DESCRIPTION

In areas where there is a reasonable risk that there will be an accidental spillage of chemicals or hydrocarbons that would be detrimental to the environment, paving shall comprise reinforced concrete. The paved areas are divided into slabs separated by flexible joints to allow for lateral movement. Where chemical and hydrocarbon spillages described above may occur, these joints shall be designed to be impermeable.

In areas where light-duty paving is sufficient and there is a low probability of a spillage, tiles, bricks or flexible pavements may be acceptable. The pavement may rest directly on the sub-grade prepared during the site preparation. However, a sub-base to support the paving may be required depending on the suitability of the sub-grade.

3.3 FUNCTIONAL REQUIREMENTS

Functional requirements for paving areas are:

- provision of reliable and easy access of personnel and equipment for construction, operation and maintenance;
- provision of a foundation base for light equipment, such as sheds, racks, ladders, stairs, pipe supports, etc.;
- protection of the soil and the ground water from contamination resulting from a spillage of chemicals or hydrocarbons;
- prevention of erosion of the soil;
- routing of spillages (liquids or gas) to sumps and drains.

3.4 DESIGN AND MATERIALS

3.4.1 Paving classification

Paving is classified as follows:

- light-duty paving;
- heavy-duty paving;
- special paving.

3.4.2 General

Designs using light- or heavy-duty paving are preferred. If the functional requirements specified in (3.3) cannot be satisfied with the use of light- or heavy-duty paving, special paving shall be designed to the Principal’s specification. For transition slabs, see (2.4.11).

3.4.3 Site preparation interface

The sub-grade and sub-base shall be prepared in accordance with (2.4.2). Reference is made to DEP 34.11.00.11-Gen. and DEP 34.11.00.12-Gen.

3.4.4 Calculation method

Paving slabs shall be designed in accordance with accepted methods, which may include the modified Westergaard's formulae. The minimum axle loads are defined in Figure 2.2.

A modulus of sub-grade reaction is normally used to represent all soil supporting the concrete paving slabs. To determine or to verify the modulus, the standard plate bearing test, with a 762 mm (30 in.) diameter plate, shall be used. For other diameters or methods, the results shall be corrected to obtain a value comparable with the standard method.

Steel wire fabric reinforcement shall be used to prevent uncontrolled cracking due to stresses resulting from such effects as shrinkage or thermal variations. It shall be assumed that the reinforcement mesh does not contribute to the tensile strength of the pavement slab.

The design of the wire mesh shall take into account the maximum friction forces due to sliding. The forces are dependent on slab base friction and weight of the slab only.

In addition to determining the modulus of sub-grade reaction, the consequences of short- and long-term settlements and differential settlements shall also be taken into consideration.

Locations of the lateral movement joints shall be selected similarly to, and together with, those of slab reinforcements to prevent uncontrolled cracking as a result of shrinkage, thermal variations, natural changes in moisture content and other conditions. Tensile strain of up to 0.015 % shall be allowed.

Special paving may be required to support heavier loads or to satisfy additional requirements. The design principles shall be the same as described above. Alternative methods shall require the approval of the Principal.

3.4.5 Standard light- and heavy-duty paving

Schematic drawings showing typical sections of standard paving are presented in Figure 3.1. Details of heavy and light duty pavements and footpaths are presented in Standard Drawings S19.003, S19.004 and S19.005. As noted in (3.2), tiles, bricks or flexible pavements may be acceptable for the design of light-duty pavements.

Where concrete is required in the construction of standard light- and heavy-duty paving, a concrete grade for reinforced concrete structures as defined in DEP 34.19.20.31-Gen. shall be specified as a minimum. The design shall take into consideration the climatic and chemical conditions. The concrete mix design shall be compliant with DEP 34.19.20.31-Gen.

The thickness of the light-duty paving shall be at least 100 mm (4 in) and a minimum reinforcement of one layer of 7 mm (1/4 in) steel wire spaced at a pitch of 200 mm x 200 mm (8 in x 8 in) shall be specified. The reinforcement material shall be compliant with material A 193 in accordance with BS 4483 or equivalent.

The thickness of the heavy-duty paving shall be at least 150 mm (6 in) and a minimum reinforcement of two layers of 7 mm (1/4 in) steel wire spaced at a pitch of 200 mm x 200 mm (8 in x 8 in) shall be specified. The reinforcement material shall be compliant with material A 193 in accordance with BS 4483 or equivalent.

Steel reinforcement shall have a minimum cover of 40 mm (1-½ in) and if only one layer is required this shall be placed at 0.6 times the slab thickness above the bottom of the slab. In aggressive environments that may result from climatic or chemical conditions, an increased thickness of cover may be required. The proposed design shall be submitted to the Principal for review.

Brick paving shall be installed in stretcher bond or fish bone bond. Bricks shall be concrete and manufactured according to EN-1338 or according an equivalent standard approved by the Principal. The sub base of brick paving shall consist of one layer of sand with a minimum thickness of 50 mm. Connections to existing paving shall be laid in similar bond as existing, possible occurring level differences shall be solved by gentle sloping. Linear level differences shall be connected with a maximum slope of 1:20 and in transverse direction with a maximum slope of 1:40.

3.4.6 Construction of lateral movement joints

Flexible lateral movement joints shall be provided between two adjacent slabs to prevent ingress of hydrocarbons and chemicals. In general, no dowels are required and the joint comprises a gap of sufficient width to cater for local temperature differentials. The joint shall be filled with elastic filler and sealed with a chemical and hydrocarbon resistant sealant.

The same method of construction shall be adopted for joints between paving and catchment basins, channels and sumps.

Lateral movement joints shall be spaced not more than 10 m (33 ft) apart. The maximum slab size shall be 20 m x 20 m (66 ft x 66 ft). To minimise the opening of lateral movement joints due to shrinkage of concrete during curing, alternate slabs shall be initially cast and cured. On completion of curing, the remaining slabs can be cast to fill in the spaces between the slabs.

3.4.7 Drainage considerations

Catchment areas and gradients of paving for certain on-plot and off-plot applications are described in DEP 34.14.20.31-Gen. Where guidance is not covered by the referenced standard, the following minimum gradients shall apply:

Paving area	Paving gradient
LNG	1 in 200
LPG spheres	1 in 50
Furnaces	1 in 50

3.5 CONSTRUCTION

The sub-base shall have a minimum thickness of 250 mm (10 in) and shall be made of free draining, well graded, granular material and shall be uniformly compacted to achieve the design requirements.

Waterproof building paper, polyethylene sheeting or equivalent material shall be laid before the reinforcement is placed to prevent absorption of water from the concrete into the formation layer. Prior to pouring the concrete, the surface shall be thoroughly cleaned of loose or other deleterious material.

The concrete shall be thoroughly densified applying a vibrator or other suitable equipment.

After pouring and finishing, the concrete shall be cured for at least seven days. To facilitate the curing process, a curing compound may be applied along with an impermeable sheet, damp fabric, or wetted sand cover during the curing period.

The construction materials shall, as a minimum, comply with the requirements listed below:

- complete mesh square for reinforcement;
- polythene sheet or waterproof building paper, compliant with BS 1521;
- reinforcement wire mesh shall be continuous through lateral movement construction joints.

4. SURFACING OF UNPAVED AREAS

4.1 DETAILED SCOPE

This section describes the minimum technical requirements for the surfacing of those areas within the site that do not require paving.

These areas are generally limited to those parts of the site where no hydrocarbon or chemical spill is expected to occur, and where the surface normally does not need to support loads other than pedestrians.

4.2 FUNCTIONAL REQUIREMENTS

Surfacing of unpaved areas shall meet the following requirements:

- prevent soil erosion by wind and water;
- be able to support pedestrians and light vehicles, including 4-wheel drive vehicles;
- repress undesirable vegetation.

4.3 DESIGN

Surface materials shall be durable and shall have minimal maintenance requirements. In the event of fire, they should be inert or self-extinguishing. The ground water shall not be polluted and, if necessary, a drainage system shall be provided for.

Unpaved areas may be classified into 2 categories as defined below:

1. Low fire hazard, which may include the following areas:
    - neutral zone between fences;
    - unused off-plot areas;
    - off-plot pipe racks and pipe tracks containing continuously welded piping with no flanged connections.

    If grass is specified, preference shall be given to simultaneous use of multiple slow growing varieties. If there is an acceptable safety distance to the nearest fire hazard area, the Principal may approve other forms of vegetation.

    For gardens and landscaping areas, the Principal shall specify requirements.

2. Fire hazard, which may include the following areas:
    - flare and open fire areas;
    - off-plot pipe racks and pipe tracks which contain flanged connections, valves or sampling points, and include manifolds;
    - areas around processing units.

    Pipe tracks shall have a 75 mm (3 in) thick gravel cover.

    In all on-plot areas, gravel, or other inert material is required in a layer thickness of at least 50 mm (2 in). Measures shall be taken to minimise the growth of vegetation.

    Off-plot areas should be covered with simultaneous use of multiple slow growing varieties of grass. Other types of surfacing, including gravel, shall require the Principal's approval.

    Gravel 16/32 shall be used.

4.4 MATERIALS

Inert materials as mentioned in (4.3) shall be durable under local conditions. Crushed, sound rock, coarse gravel, sand-cement mixes and blast furnace slag are generally acceptable.

5. CABLE TRENCHES

Note: This section is limited to general civil engineering minimum requirements only.

DEP 33.64.10.33-Gen., DEP 32.37.20.10-Gen. and DEP 33.64.10.10-Gen. describe electrical and instrumentation requirements for cable trenches.

Cables laid in cable trenches shall be at least 600 mm (24 in) below ground level. The depth of the trench may be increased in order to accommodate up to 3 layers of cables. Trenches shall be filled with graded, non-angular, well draining, compacted sand free of sharp particles, e.g. sieved. The design depth of and backfill materials for cable trenches shall take into consideration the detrimental effects of frost in geographical regions, or locations at the plant where sub-zero temperatures can be expected. Salty sand shall not be used unless specifically approved by the Principal.

In paved and unpaved areas, cables shall be shielded by cable tiles at a depth of at least 350 mm (14 in) below final grade. Under concrete paving no cable tiles are required. The location of trenches shall be clearly indicated on the surface, by colour code in paved areas and by signs in unpaved areas.

In paved areas, trenches that need to remain accessible shall be covered with removable concrete panels. Panels shall be designed to the same standard as the surrounding paving.

If access to trenches in paved areas is not required, the permanent paving should be continuous.

Instrument trenches shall always cross at an angle of 90 degrees with power and lighting cables. For the upper trench, a bridge comprising a concrete bottom slab supported on the walls of the lower trench shall be constructed to facilitate excavation of the lower cable trench.

Separation distances between instrument and power and lighting cables shall be as per DEP 33.64.10.33-Gen., DEP 32.37.20.10-Gen. as per DEP.

The bottom of trenches should be above the permanent ground water table. In designing the layout and depths of cable trenches, consideration shall be given to the design requirements of the drainage system

The cables shall have a minimum distance of 0.3 m (1 ft) to any buried pipelines. In hot lines the pipe shall be insulated to limit the temperature at the outside to 60 °C (140°F) maximum, reference is made to DEP 33.64.10.10-Gen.

To allow efficient design and installation of underground utilities, at the beginning of the works an underground elevation protocol specifying depth and installation sequence of underground utilities, including cables, shall be specified. Cables should cross underneath buried pipelines.

Details of electrical and instrument cable trenches and routes are presented in Standard Drawings S19.001 and S19.002.

6. EROSION PROTECTION OF SLOPES INCLUDING, EMBANKMENTS, DITCHES AND OPEN DRAINS

6.1 DETAILED SCOPE

The scope of this section is limited to erosion protection aspects.

The stability, settlement behaviour and construction of slopes, bund walls and embankments are covered in DEP 34.11.00.11-Gen. and DEP 34.11.00.12-Gen.

For ditches and open drains, only the material, design, and construction aspects of the protective surface layer are considered.

Design and capacity requirements for drainage systems are covered in DEP 34.14.20.31-Gen.

6.2 GENERAL DESCRIPTION

The purpose of erosion protection is to prevent the process of weathering and transport of solids (sediment, soil, rock and other particles).

A filter fabric shall be applied if a wash out of sand or sandy material is expected through the erosion protection layer, e.g., through joints in brickwork and slabs, cracks in the protection layer, etc. The possibility of cracking should also be considered.

6.3 FUNCTIONAL REQUIREMENTS

Slopes for bund walls, cuts, embankments, open drains and ditches, for example, shall be protected against erosion and subsequent damage, and failure on a micro scale as a consequence of wind, water, and spillages. In the selection of the slope protection system, other aspects shall be taken into consideration to ensure a durable and low maintenance solution is adopted. Selection shall take into consideration activities that may occur in the close vicinity of the slope and could have a detrimental effect, e.g., temporary loading from vehicles, storage of materials, or vibration.

For guidance on protection against failure on a macro scale (slope stability) reference is made to DEP 34.11.00.12-Gen.

In areas where there is a reasonable risk that there will be an accidental spillage of chemicals or hydrocarbons that would be detrimental to the environment, the cover layer of open drains and ditches shall protect the subsoil against pollution. The protection measures adopted shall be selected based on a study of the level of risk. If an accidental spillage flows through the system rapidly, and a cleanup operation can be completed in a short time, then the protection system adopted may possibly include a lining that is not impermeable. In instances where the spillage cannot be cleared away rapidly, an impermeable membrane solution shall be adopted.

6.4 DESIGN

Erosion protection systems shall be selected to cope with all requirements under all operational conditions. The criteria shall be proposed to the Principal for review.

The design shall incorporate measures to prevent the sliding downwards or floatation of the erosion protection system as a result of changes in pore water pressures, variations in temperature, slumping from self-weight or other processes. The design of the slope protection shall consider the long term superficial settlement, erosion or other behavioural processes, and ensure that the slope protection system remains serviceable.

Where impermeable erosion protection systems are selected to prevent contamination of ground and groundwater, the design shall not include drainage points or weep holes. However, the design shall take into account the highest possible ground water level in combination with the minimum water level in the drain to design against slope failures due to build up of pore water pressures.

6.4.1 On-plot areas

6.4.1.1 Inside battery limits

Inside battery limits, all slopes, bund walls, embankments, open drains, and ditches shall be lined with concrete. The concrete shall be designed in accordance with DEP 34.19.20.31-Gen. Special care shall be taken in pouring these relatively thin slabs, and as a minimum the requirements of DEP 34.19.20.31-Gen. shall be followed. The minimum layer thickness of the lining concrete shall be 80 mm (3-1/8 in) and a reinforcement mesh of 7 mm (1/4 in) steel wire spaced at a pitch of 200 mm x 200 mm (8 in x 8 in) or equivalent shall be incorporated into the design.

Slabs shall have a maximum size of 5 m x 5 m (16 ft - 6 in x 16 ft - 6 in).

Where the risk of contamination is considered acceptably low, slopes higher than 1 m (3 ft - 3 in) shall be provided with drainage points or weep holes to eliminate the build up of pore water pressures. Wash out of soil through these drainage points or weep holes shall be prevented.

Erosion of soil through joints between panels shall be prevented, e.g., by the application of strips of suitable filter material. Long-term behaviour, e.g., danger of gradual clogging, shall be taken into consideration.

6.4.1.2 Outside battery limits

Grass shall not be used as erosion protection in flare areas, major pipe tracks, pump pits, or other areas where there is a requirement to minimise fire hazards.

In areas where there is unlikely to be an accidental spillage of chemicals or hydrocarbons, any erosion protection system, except grass, shall be acceptable.

For concrete panels the minimum requirements described in (6.4.1.1) shall apply.

Drainage points or weep holes as specified in (6.4.1.1) shall be provided for flexible erosion protection systems.

Where water-tight or low permeable erosion protection layers are required, the drainage aspect as discussed in (6.4.1.1) regarding drainage points or weep holes shall be considered.

In areas where there is a reasonable risk that there will be an accidental spillage of chemicals or hydrocarbons that would be detrimental to the environment, one of the following systems will be adopted:

- concrete slabs as described in (6.4.1.1);
- for major drains, an impermeable sealing membrane shall be used to at least 0.5 m (1 ft - 8 in) above design water level.

6.4.2 Off-plot areas

In all areas not covered by (6.4.1), grass is recommended.

6.5 MATERIALS

Materials shall be consistent with ASTM Volume 04.03 or the equivalent accepted by the Principal. For concrete and cement mortar, see DEP 34.19.20.31-Gen.

6.6 CONSTRUCTION

Slopes shall be compacted and trimmed before erosion protection layers are installed to prevent local failure. Proposed working methods shall be submitted to the Principal for review.

7. FENCING AND GATES

7.1 GENERAL

This section covers only the minimum requirements for permanent fences and gates. It does not cover special security measures that may be required such as cameras, security guards and other specialty fencing systems. The design of temporary fencing shall be agreed with the Principal.

The standards referenced below shall be used to supplement the requirements in (7.2) to (7.7):

- ASTM A 392;
- ASTM A 491;
- ASTM D 1557;
- ASTM F 552;
- ASTM F 567;
- ASTM F 626;
- ASTM F 668;
- ASTM F 1043;
- BS 1722.

7.2 SECURITY RISK ASSESSMENT

For green field developments and extensions of existing facilities, a security risk assessment shall be executed. The outcome of the assessment shall identify the category of risk the site is exposed to in accordance with those described below.

The fence type can be uniform for the whole facility, but where particular risk areas are identified, other fence types may be required.

The assessment shall classify the site according to one or more of the following five risk categories:

- insignificant risk - a situation with a minimum level of crime and a need for basic personal and company security precautions. Generally an effective public security infrastructure is in place;
- low risk - a situation with frequent incidents of petty crime, limited possibility of activism or terrorism, and generally good quality public security infrastructure. Basic personal and company security are required;
- medium risk - a situation with frequent incidents of crime, violence perpetrated by criminals, terrorists or guerrillas, and the possibility of small-scale internal unrest. Generally limited effectiveness of the public security infrastructure. Attention must be paid to good personal and company security precautions;
- high risk - a situation with frequent petty, serious, and organized crime. A serious terrorist or guerrilla problem, with the possibility of medium-scale internal unrest with the threat of internal armed conflict. Generally poor quality public security infrastructure. Stringent personal and company security precautions are required;
- critical risk - a situation where the security authorities no longer have control of law and order, and personnel and the company have to maintain their own highest possible levels of protective measures.

7.3 GENERAL DESCRIPTION

7.3.1 Fence types

Four examples of typical fencing are shown on Standard Drawings S13.001, S13.002, S13.003 and S13.004. The security risk in the area and the suitability of locally available materials will govern the appropriate selection of fencing for a particular site.

7.3.2 Preferred fencing system

The preferred system consists of the following components:

- all posts for line, corner and end have a buried concrete footing and inclined top;
- bracing;
- chain link fence fabric;
- barbed wire;
- tension wire;
- gates.

7.3.3 Alternatives

Alternatives to the preferred system shall be submitted to the Principal for review and may be considered provided they satisfy the requirements of (7.4) and (7.5).

7.4 FUNCTIONAL REQUIREMENTS

7.4.1 Fences

A system of fences and gates shall be erected close to and entirely within the site limits.

The system shall be designed to prevent unauthorised entry into the site by persons who are not equipped with special means to force such an entry.

The system may consist of a single or double fence as required to address the perceived risk assessed in (7.2) as a minimum of 2.4 meters height. Where determined as a requirement by Security Risk Assessment, weld mesh, expanded metal or metal palisade fencing material shall be used for perimeter fencing.

The top of the external boundary fence shall have a 45° outward inclined section of 0.7 m (2 ft - 4 in) length with anti-climb topping to 2.9 metres overall. The top of the outward inclination shall be within the property limits. Maintenance tracks or patrol roads may be required as specified by the Principal.

Anti-climb topping shall be either

(1) 3–6 strand barbed wire top-guard constructed of 9 gauge or heavier wire angled out and up at a 45 degree angle;

(2) concertina wire/razor-ribbon.

The system shall also facilitate discovery by plant security of any trespassing. A clear line of sight shall be established along the length of the fence with hills and gullies along the fence line being levelled off as far as practicable. Fences shall have a minimum number of bends.

The system shall not inhibit the natural dispersion of gas leakages on site.

Where natural elevation, gradient, or man-made structures effectively compromise the height of a 2.9 metres perimeter or critical area fence, additional fencing material shall be added to ensure an effective overall height of 2.9 metres is maintained.

Fencing should be buried at the base wherever practicable. Where not possible, gaps between the bottom of the fence and grade shall not exceed 5 cm.

Storm drains, culverts, pipelines, utility tunnels, etc. in excess of 620 square cms and which pass through or under the perimeter fence, shall be fitted with additional security screening or bars to preclude intrusion.

7.4.2 Gates

Gates shall be provided to satisfy operational requirements. However, the number of gates provided to meet these requirements shall be kept to a minimum.

Gates shall be an integral part of the fence and shall be of a similar design with respect to height and strength, etc.

7.5 DESIGN REQUIREMENTS

7.5.1 General

Prior to the erection of fences, site preparation shall fulfil the following requirements:

- embankments, which may be required under (7.4.1), shall not extend beyond the site boundary;
- embankments shall be stable and have a slope of maximum 1 vertical to 1.5 horizontal. Adequate erosion protection shall be applied;
- embankment fill material shall be placed at a minimum dry density of 85 % modified AASHTO (ASTM D 1557 or AASHTO T180);
- embankments shall have a minimum shoulder of 0.5 m (1 ft - 8 in) between the fence foundation and the outside embankment slope;
- embankment slopes, etc., resulting from site preparation works shall allow a clear view from the patrol road or maintenance track towards the area outside the fence.

Exceptions shall be submitted to the Principal for review. The fence design shall comply with BS 1722-10.

7.5.2 Preferred fence design

The fence shall be designed to withstand any load that may occur (such as gale force wind loads, thermal movement from extreme climatic conditions, etc.).

The vertical part of the fence and gates shall be a minimum of 2 m (6 ft - 6 in) tall and the gap between the fence and the ground shall at no place exceed 0.1 m (4 in). The extent of support at the base of the fence shall depend on the local environmental and climatic conditions as well as the requirements necessary to address the perceived risk assessed in (7.2) and other considerations including national requirements. Ground beams and anchors may be considered to prevent the bottom of the fence from lifting up.

The concrete footing of a fence post shall extend at least 0.6 m (2 ft) into the ground, and shall have at least the same weight as the post with a minimum of 25 kg (55 lb).

Both cast in situ and precast concrete footings are acceptable.

Metal parts, except the chain link fence fabric, shall not be in contact with the soil.

Gates shall be of the swinging type unless otherwise specified by the Principal.

7.5.3 Drainage crossings

At crossings of the fence with water courses or other drainage channels, a removable bar screen shall be installed for security reasons.

7.6 MATERIALS

The material choice shall be based on local availability, minimum maintenance and low capital costs. Galvanized or aluminium-coated steel, aluminium and concrete are generally acceptable. PVC-coated chain link fence fabric and tension wire may be used.

Materials shall be durable in view of the climatic conditions, soil conditions and plant atmosphere.

All materials shall be compatible and not lead to contact corrosion when assembled.

7.7 TIMING OF CONSTRUCTION

The construction of permanent fences and gates shall precede all other construction activities unless otherwise agreed by the Principal. In the latter case a temporary fencing system shall be considered.

8. MAINTENANCE

A maintenance cycle to ensure assets remain serviceable for the life of the site shall be implemented for all civil engineering infrastructures. The designer and site operator shall develop a master plan that shall incorporate a system of reporting results of both routine and irregular inspections of the assets. The system shall catalogue the condition of assets and criteria shall be established which shall trigger the requirement for repair or renewal. This system shall ensure the assets remain serviceable and allow the operating unit to plan for maintenance that will ensure maintenance costs can be included in annual budget plans and kept to a minimum.

9. REFERENCES

In this DEP reference is made to the following publications:

NOTES: 1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

2. The DEPs and most referenced external standards are available for Shell users on the Shell Wide Web (SWW) at address http://sww.shell.com/standards.

SHELL STANDARDS

Standard drawings index	DEP 00.00.06.06-Gen.
Instrument signal lines	DEP 32.37.20.10-Gen.
Electrical engineering design	DEP 33.64.10.10-Gen.
Electromagnetic compatibility (EMC)	DEP 33.64.10.33-Gen.
Site preparation and earthworks including tank foundations and tank farms	DEP 34.11.00.11-Gen.
Geotechnical and foundation engineering - Onshore	DEP 34.11.00.12-Gen.
Drainage and primary treatment facilities	DEP 34.14.20.31-Gen.
Reinforced concrete structures	DEP 34.19.20.31-Gen.
Layout of onshore facilities	DEP 80.00.10.11-Gen.
Fire-fighting vehicles and fire stations	DEP 80.47.10.33-Gen.

STANDARD DRAWINGS

Fence construction type “A” with reinforced concrete posts	S13.001
Fence construction type “B” with tee posts and horizontal tubing	S13.002
Fence construction type “C” with tee posts	S13.003
Fence construction type “D” with tubular posts	S13.004
Typical sections of heavy duty roads (flexible and rigid paving)	S13.005
Typical sections of plant roads (heavy duty and rigid paving)	S13.006
Typical sections of light duty roads and patrol roads (flexible and rigid paving)	S13.007
Electrical and instrument cable trenches in concrete paved areas	S19.001
Cable routing in unpaved, brick-paved or tiled areas and crossing roads	S19.002
Typical lay-out plans of paving	S19.003
Light-duty concrete paving and footpaths - typical details	S19.004
Heavy-duty concrete paving - typical details	S19.005

AMERICAN STANDARDS

Standard method of test for moisture-density relations of soils using a 4.54-kg (10-lb) rammer and a 457-mm (18-in.) drop	AASHTO T180
Guide for design of pavement structures	AASHTO GDPS-4

Issued by:
American Association of
State Highway and Transportation Officials
444N Capitol Street, NW, Suite 225
Washington DC, 20001

Standard specification for zinc-coated steel chain-link fence fabric	ASTM A 392
Standard specification for aluminium-coated steel chain-link fence fabric	ASTM A 491
Standard test methods for laboratory compaction characteristics of soil using modified effort (56,000 ft-lbf/ft3 (2,700 kn-m/m3))	ASTM D 1557
Standard terminology relating to chain link fencing	ASTM F 552

Standard practice for installation of chain-link fence	ASTM F 567
Standard specification for fence fittings	ASTM F 626
Standard Specification for Polyvinyl Chloride (PVC), Polyolefin and Other Polymer - Coated Steel Chain Link Fence Fabric	ASTM F 668
Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework	ASTM F 1043
Road and paving materials; vehicle-pavement systems	ASTM Volume 04.03

Issued by:

American Society for Testing and Materials

100 Barr Harbor Drive

West Conshohocken, PA 19428-2959
USA

BRITISH STANDARDS

Specification for waterproof building papers	BS 1521
Fences	BS 1722
Part 10: Specification for anti-intruder fences in chain link and welded mesh	BS 1722-10
Steel fabric for the reinforcement of concrete - Specification	BS 4483

Issued by:

British Standards Institution

389 Chiswick High Road, London W4 4AL

United Kingdom

EUROPEAN STANDARDS

Concrete paving blocks - Requirements and test methods	EN 1338:2003

Issued by:

European Committee for Standardization

Rue de Strassart 36, B-1050, Brussels

Belgium